

lukewilsen45@gmail.com
(978) 489-4234
Concord, MA, 01742

Luke Wilsen

GitHub
LinkedIn

EXPERIENCE

Analog Devices Inc. High-performance analog & digital semiconductors.

Wilmington, MA

Associate Software Engineer, Developer Experience (DevEx)

Feb 2026 – Present

- Engineered and containerized an end-to-end telemetry pipeline with Docker Compose and Streamlit, automating cross-ecosystem tracking of Linux kernel contributions and open-source developer activity.
- Partnered with Linux kernel maintainers across an 8-week validation cycle to define 20 core KPIs, establishing a framework to understand and quantify open-source value creation.
- Deployed an automated Claude Code telemetry pipeline, delivering actionable utilization data to guide enterprise AI tool adoption.

Iron Eagle X Defense Technology & AI solutions

Tampa, FL

Data Science Intern

Jan 2024 – Dec 2024

- Engineered and productionized end-to-end ML pipelines, integrating Flask APIs and Docker optimized for cloud-based scalability.
- Utilized CI/CD practices to automate model testing and deployment, ensuring 100% reproducibility.

Prapela Pediatric Medical Device Company

Biddeford, ME

Statistical Consultant

Sep 2023 – Feb 2025

- Advised on FDA De Novo submission that secured clearance for a pediatric device with Breakthrough Designation.
- Performed statistical analysis (Python, R) of sensor data to validate stimulation equivalency across clinical and production devices.

EDUCATION

University of Massachusetts Amherst

Amherst, MA

Master of Science in Biostatistics, GPA 4.0

May 2025

Relevant Courses: Multivariate Calculus, Linear Algebra I&II, Bayesian Modeling, Causal Inference, Statistical Methods for Data Science

University of Massachusetts Amherst

Amherst, MA

Bachelor of Science in Public Health, GPA 3.7

May 2024

PROJECTS

Covid Outcome Bayesian Modeling

- * Developed Bayesian hierarchical GLMs to quantify hospitalization risk from COVID-19, using spatial statistics, hypothesis testing, and causal inference to uncover risk disparities by geography and race.

Length of Stay and Disease Severity Prediction

- * Built a combined cohort regression model to predict disease severity and length of stay using synthetic claims data that achieved MAE of 0.149 days outperforming existing literature based models by 92.5%.

SKILLS

Programming: Python, R, SQL, Bash, Pandas, NumPy

Platforms & Data Infrastructure: GitHub, SQL, Linux HPC, Jenkins, Prometheus, OTEL, Cloud Environments (AWS — familiar)

Deployment & Tools: Amazon Web Services, Docker Compose, Git, Flask, FastAPI, Streamlit, Tableau, Power BI, CI/CD, Version Control

Statistical Methods: Cohort Analysis, Bayesian Modeling, GLMs, Hypothesis Testing, Time Series, Spatial Analysis, Causal Methods (IPW, matching, experiment design), A/B Testing, Experimental Design

ML & Data Science: Scikit-learn, TensorFlow, PyTorch, Deep Learning, NLP, LLMs, Classification, Regression, Clustering, Time-Series Analysis, Feature Engineering, Developer Experience (DevEx)